Game Theory in the World

Essay Writing Guideline



What Your Essay Must Do

Choose a strategic phenomenon, ask a focused question, and use a game to answer it.

1. **Explain the setting:** give the essential facts and identify your source or observation.
2. **Build a game:** specify and justify the players, choices, timing, information, and payoffs.
3. **Define the solution:** specify each player's strategies and state the equilibrium conditions for your game, using Nash equilibrium or another solution concept learned in class.
4. **Solve it:** show the reasoning that establishes the equilibria.
5. **Discuss implications:** answer the question, examine one meaningful change, and explain a specific limitation.

Due Date and Format

Due: Wednesday, December 16, 2026, by 2:20 PM. Worth **10%** of the course grade; graded out of 100 points.

1,000–1,500 words of prose, about 4–6 double-spaced pages in 12pt font with standard margins. Follow the word limit; the page count is only an estimate. Figures, matrices, displayed equations, captions, and references are excluded.

Use **one title and no section headings**. Write the essay as continuous prose organized into paragraphs, not as a list of answers.

Below the title, include your name, student ID, and the word count of the prose. Keep essential explanations in the essay, not captions or appendices.

Choose a Manageable Strategic Question

Use a news article, an everyday experience, a movie, or a television show. Published sources are optional; cite any you use. Identify personal observations as such. Fiction supports an interpretation of the story, not an empirical claim.

Ask, for example, why rival firms keep discounting, why a threat fails to deter entry, or why people wait for others to contribute.

Choose an interaction in which each player's best choice depends on what others do. A well-justified two-player, two-action game can earn full credit. Complexity does not necessarily earn higher credit.

Define the Game Completely

1. **Players and choices:** who decides, and what can each player do? Justify treating a group as one player.

2. **Timing and information:** who moves when, and what do they know? Specify uncertainty and probabilities if needed.

3. **Payoffs:** give every outcome's payoffs or a formula that determines them, and explain what they represent.

Present a payoff matrix, game tree, or precise mathematical specification. A verbal description alone is insufficient.

Justify Assumptions and Payoffs

Separate **facts from your source or observations** from **assumptions you introduce**. Explain why your simplifications capture the strategic issue.

You may invent reasonable payoff numbers, but label them as illustrative. Explain how the payoffs represent the players' incentives; do not choose numbers merely to force a desired result.

To find best responses to pure strategies, rankings of payoffs suffice. Against mixed strategies, expected utility depends on differences in payoffs as well as rankings. Explain what your numbers mean; do not present invented rankings as measured profits.

Define the Solution, Then Solve the Game

Specify each player's strategies, including actions at every information set in a sequential game. State equilibrium conditions using your game's players, strategies, and payoffs; a generic textbook definition is insufficient.

Then find the strategies that satisfy those conditions and show why.

- **Simultaneous choices:** derive best responses and check deviations.
- **Mixed strategies:** derive probabilities from indifference and check excluded actions.
- **Sequential or incomplete-information games:** show sequential reasoning or beliefs and expected payoffs, as relevant.

Find all equilibria under your chosen concept, including mixed strategies. You may restrict attention to pure strategies if you justify the restriction and discuss what it leaves out. Find all equilibria within that scope.

Interpret and Test the Result

Translate the equilibrium into behavior in your setting. Explain how the incentives answer your original question, and compare the prediction with the source or observation.

Change **one** payoff, assumption about timing or information, or available action. State what changes and what stays fixed. Compare the old and new incentives, and explain whether the equilibrium changes.

Explain **one specific limitation** and how it could matter. "The real world is more complicated" is insufficient. Agreement with the story does not prove your assumptions; a carefully explained mismatch can also be informative.

Example: Advertising by Rival Shops

Two shops simultaneously choose whether to advertise. Table 1 gives illustrative profits, with Shop 1's payoff first.

	Shop 2: Advertise	Shop 2: Do not
Shop 1: Advertise	$(2, 2)$	$(5, 1)$
Shop 1: Do not	$(1, 5)$	$(4, 4)$

Table 1. Illustrative profits for the two shops.

Each shop's pure strategies are to advertise or not. In a Nash equilibrium, neither shop gains by switching while the other's choice stays fixed. Against Shop 2's advertising, Shop 1 compares 2 from advertising with 1 from not advertising.

For each shop, advertising yields $2 > 1$ if its rival advertises and $5 > 4$ otherwise. Thus, both advertise in the unique Nash equilibrium, although both would earn more if neither did.

From the Example to an Implication

Each shop now pays an additional cost $c \geq 0$ if it advertises. Actions, timing, and all other payoffs stay fixed. Against either rival action, the gain from advertising becomes $1 - c$.

- If $c < 1$, advertising remains strictly dominant for both shops.
- If $c > 1$, not advertising becomes strictly dominant for both.
- If $c = 1$, both are indifferent, so every strategy profile is a Nash equilibrium.

This identifies the cost at which the shops' incentives change. It does not establish whether restricting advertising is desirable: the model omits benefits to consumers.

Draft Early and Revise

Write while you build the game. Explaining a payoff or a best response in words can reveal a missing assumption or an error in the model. Revise the model when the explanation exposes a problem.

Make an outline of claims, such as "Advertising is each shop's best response," rather than empty labels such as "Analysis." The outline can change as you work.

Read the draft aloud. Rewrite sentences that are awkward to say or hard to follow. Then ask a classmate to identify your question, main result, and the reason for it; revise where the reader loses the argument.

Write One Continuous Argument

Write for a classmate who knows the concepts taught in class but has not read your source. Explain the setting and your assumptions without retelling the course.

Use a specific title, such as "Why Rival Shops Keep Advertising." Begin with the specific situation, strategic question, and your answer. Avoid sweeping openings such as "Throughout history, humanity..." or "In today's society..." and announcements such as "This essay will discuss game theory."

Move from the setting and assumptions to the game. Specify strategies and state equilibrium conditions for that game, then solve it and discuss implications. Each paragraph should develop one point and make it clear near the beginning, after a brief connecting sentence if needed. Develop the point, then move on; do not add a closing sentence that merely repeats it.

Connected Sentences, Consistent Terms

Connect each sentence to the preceding idea or the paragraph's central subject. Repeating a useful term can clarify the connection; you need not build a rigid chain from one sentence to the next.

Example. Advertising attracts customers from the rival shop. Those customers make advertising profitable even when the rival also advertises. So neither shop wants to stop advertising on its own.

Keep names and terms consistent: do not switch from "Shop 1" to "the incumbent" merely for variety.

Use "however" or "therefore" when the relationship calls for it. Adding a transition word cannot repair a missing logical connection.

Sentence-Level Style

- Prefer active voice when it clarifies who acts: "Shop 1 advertises." Passive voice is useful when it keeps the reader's attention on the right subject.
- Use familiar words and concrete comparisons: "deviating raises its payoff from 1 to 2," not "deviating is obviously better."
- Cut generic openings, needless qualifiers, and repeated summaries. Keep the assumptions and reasoning the reader needs.
- Use notation only when it helps. Define each symbol and its economic meaning when needed; avoid inventing a symbol for something mentioned only once.

Assumptions, Results, and Interpretations

Make your meaning unambiguous. Identify whose payoff changes, the alternative being compared, and the assumptions that apply.

- **Assumption:** "I assume that the shops choose simultaneously." Explain which feature of the setting motivates this assumption.
- **Result:** "Both shops advertise in equilibrium." Establish this with comparisons of payoffs; an assumed incentive is not a proved result.
- **Interpretation:** "This may explain the spending described in the article." Explain the connection without claiming the model proves what caused it.

Positive versus Normative Claims

Positive claims describe what happens: "Under these assumptions, each shop advertises." **Normative claims** concern what should happen: "Advertising should be restricted."

Keep these separate. A Nash equilibrium need not be efficient, fair, or desirable. If you recommend a policy, state whose welfare matters and the tradeoffs your criterion includes.

You need not recommend a policy. You can earn full credit for implications through prediction, interpretation, a change in assumptions, and limitations.

Figures and Mathematics

Introduce each table or figure in the preceding prose. Number it, give it a short caption, place it nearby, and refer to it explicitly.

- **Matrices:** label players and strategies, and state whose payoff appears first.
- **Trees:** label movers, actions, payoffs, information sets, and chance probabilities where relevant.
- **Equations:** show enough steps to verify the solution. Use prose between equations to explain what each comparison establishes.

Hand-drawn figures are fine. Insert a clear scan or photograph of your matrix or tree. Make every label readable and consistent with the text.

Cite Sources and Use Assistance Critically

If you use an article, identify it in the opening paragraph: publication, title, author if listed, and date. For fiction, identify the work and scene; for personal experience, describe the context. List any published sources you use in a consistent reference list with links.

Paraphrase in your own words and attribute borrowed ideas; quote sparingly. Concepts taught in class need no citation, but specific borrowed models and arguments do.

You may discuss the essay with classmates and use generative AI. Verify all reasoning, wording, and citations; you must understand and defend everything submitted. Follow the syllabus on plagiarism and reuse of previous work.

Grading Rubric

The six categories below sum to 100 points.

Category	**Points**
Question and context	10
Game and assumptions	25
Solution and reasoning	25
Implications and limitations	20
Writing craft	15
Presentation and attribution	5

Context and Model

Question and Context: 10 Points

- **Full credit:** Focused strategic question; accurate, concise context; main source or observation clearly identified.
- **Common weaknesses:** Broad topic without a question; excessive summary; unclear strategic interaction.

Game and Assumptions: 25 Points

- **Full credit:** Complete game in a matrix, tree, or mathematical specification; timing and information clear; payoffs and assumptions justified; facts and assumptions distinguished.
- **Common weaknesses:** Missing players, choices, or outcomes; unexplained numbers; timing inconsistent with the story.

Solution and Implications

Solution and Reasoning: 25 Points

- **Full credit:** Strategies and equilibrium conditions specified for the student's game; all equilibria derived and verified under the chosen concept. Any restriction to pure strategies is justified and its limitations discussed.
- **Common weaknesses:** Generic definition without applying it to the game; incomplete strategies; equilibrium asserted without work; profitable deviations or equilibria within the stated scope overlooked.

Implications and Limitations: 20 Points

- **Full credit:** Answers the question; connects prediction to the setting; analyzes one change while stating what stays fixed; explains a specific limitation; separates normative claims when relevant.
- **Common weaknesses:** Repeats the result without interpretation; vague extension; generic caveat; unsupported claim about welfare.

Writing and Presentation

Writing Craft: 15 Points

- **Full credit:** Specific title; purposeful paragraphs; connected sentences; consistent terms; precise, concise prose; mathematics explained in words.
- **Common weaknesses:** Disconnected paragraphs; vague claims; needless repetition; unexplained notation.

Presentation and Attribution: 5 Points

- **Full credit:** Required length and identifying information; one title, no section headings; readable, labeled figures when used; any sources and borrowed ideas properly cited.
- **Common weaknesses:** Missing labels or source details; unreadable figures; failure to follow format or length requirements.

Levels of Performance

Each band describes the share of a category's available points earned.

90–100%	Excellent: precise, persuasive; no substantive gaps.
80–89%	Strong: meets criteria well; minor weaknesses.
70–79%	Satisfactory: essentials present; uneven explanation.
60–69%	Weak: significant gaps or errors.
Below 60%	Poor or incomplete: substantially misses criteria.

For example, 80% of the 25 points for the solution is 20. Correct steps earn partial credit even if the final conclusion is incorrect; a missing component earns zero. An error may affect both the solution and its implications.

Before You Submit

- Does the opening state a focused question and main answer?
- Have I distinguished facts, assumptions, results, and interpretation?
- Have I specified strategies and equilibrium conditions for my game, then derived and verified the equilibria?
- When testing a change, have I stated what stays fixed and explained the implications?
- Have I discussed one specific limitation, including what a restriction to pure strategies leaves out if I impose one?
- Does each paragraph develop one point? Are sentences connected and terms consistent? Have I cut needless repetition?
- Have I read the draft aloud, separated predictions from judgments, checked sources and figures, and followed the length and format requirements?

Tools and Further Reading

- **Microsoft Word:** choose Insert → Equation (Alt+= on Windows, Control+= on Mac). Use the equation tools for fractions, subscripts, and symbols. Word help.
- **LibreOffice Writer:** choose Insert → OLE Object → Formula Object to open Math. In older versions, look under Insert → Object. LibreOffice help.
- **LaTeX:** optional. Overleaf is a common online platform for writing in LaTeX. See its beginner's guide and tutorials to get started.

Further reading: McCloskey, *Economical Writing*, 3rd ed. (2019); Strunk and White, *The Elements of Style*; Harvey, *Elements of the Academic Essay*.